

Clustering Breast Cancer Patients Based on Proteome

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Abstract

In the ongoing effort to make cancer treatments more specific, diagnostic tools must be developed to characterize patients. One such tool is a set of 50 genes (PAM50), used for grouping breast cancer patients based on microarrays. In the present work, we analyze protein expression levels in 77 individuals with breast cancer to assess the possibilities for using proteomics, rather than transcriptomics, to group these patients. Three methods were used to cluster detected proteins that correspond to PAM50 genes, as well as all detected proteins: k-means, fuzzy and hierarchical clustering. For k-means and hierarchical, two different distance measures (squared Euclidean and cosine) were used; for fuzzy clustering, two values were used as fuzzy factor (1.1 and 2.0). The quality of the resulting clusterings was assessed using Silhouette analysis and compared to results from mRNA-based grouping of the same patients by Adjusted Rand Indices (ARI).

None of our clusterings are convincingly similar to the pre-existing grouping by microarray. However, those based on the reduced dataset containing PAM50 proteins only, provide better clusterings than on all the proteins, suggesting that it is more useful to use these than it is to use a full proteome for diagnosis. Silhouette scores of PAM50 clusterings are also much better than those for the entire dataset, further supporting the idea that too much unrelated variation is present in the larger dataset. Additional work is needed to improve the quality of both the dataset and the subsequent clustering, for example by increasing sample size and choosing different combinations of proteins to use.

1 Introduction

Breast cancer is considered to be one of the most invasive cancers among women in both developed and developing countries. Establishing a proper cancer diagnosis is the first critical step in disease management and leads to optimized patient treatment [1]. Currently four different breast cancer types have been described according to the Prediction Analysis Microarray 50 (PAM50) set of proteins. The PAM50 set was compiled from genome-wide expression profiling from microarray datasets and progressed to a PCR-based test with a curated list of 50 genes [6]. Although different types of breast cancer have been well defined using the PAM50 dataset, these genes only represent the genomic side (mRNA aspect) of the disease. Subgroups defined on the proteome level have not been so well characterized. Proteins drive cellular functions - more directly than does mRNA - and are also the most active and drug-treatable cellular components [3]. Therefore subgroups defined based on proteomics are needed to get a better understanding of the disease and find new therapies.

The aim of this project is to see how breast cancer patients cluster together using K-means, hierarchical and fuzzy clustering given their protein expression levels. Since four subtypes have been defined by PAM50, we are interested to see if the protein data will also divide the patients into four subgroups and how similar the clusters are to those based on PAM50-mRNA classification. We replicate a part of the study by Mertins *et al.* (2016) [4]. The dataset used and additional details can be found on the Kaggle webpage (kaggle.com).

The analysis is divided into two parts. In the first part, the proteomic data that corresponds to the PAM50 genes was clustered. This provides insight into the potential of using PAM50 protein levels, instead of mRNA levels, to group breast cancer patients. In the second part, we aim to get a more comprehensive view on groups of patients based on their proteomes. For this purpose, clustering is performed on all protein expression levels available in the dataset.

2 Methods

2.1 Data inspection

2.1.1 Whole dataset

The protein data were collected by tandem mass spectrometry (MS/MS) in the form of a ratio which compares sample expression level to an iTRAQ reference solution. The values were already on a log2 scale, so no further scaling was necessary.

The initial dataset contains expression levels for 12 553 proteins in 83 samples. There were three biological replicates present for three proteins. We combined the values for each of these pairs of samples by averaging their values for each protein. Furthermore, three healthy individuals were removed from the dataset. This left us with data on each protein in 77 patients.

2.1.2 PAM50 proteins

Of the proteins included in the dataset, 43 are products of genes in the PAM50 set. Using a small Python script, values for these proteins were extracted from

the dataset for analysis of only PAM50 proteins.

2.2 Missing values

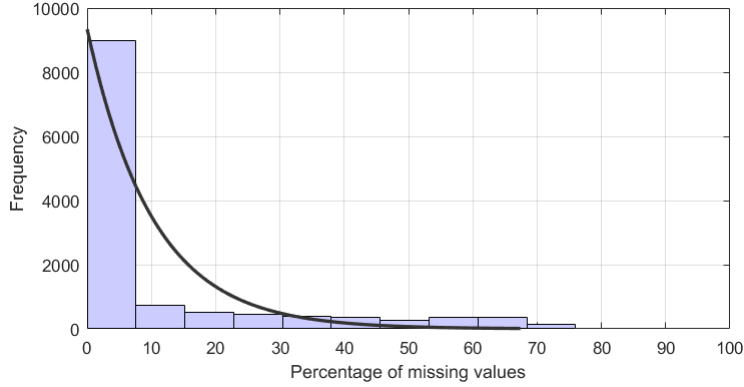
2.2.1 Whole dataset

Due to the fact that MS/MS may not be able to quantify all proteins in each sample, the dataset contained many missing values. Figure 1 shows that the majority of data points had less than 10% of values missing. To circumvent problems that could arise as a result of using variables with many "fake" (imputed) values, those proteins with values missing for more than 10% of patients were removed. This reduced the size of the dataset to 77x9313 (patients and proteins, respectively). For missing values of the remaining proteins we used k-nearest Neighbor imputation with Euclidean distance. Choosing the right value for K is a topic of debate, but following to Hassanat *et al.* (2014) [2] we decided to use $k = 8$ based on the square root of the number of patients.

2.2.2 PAM50 proteins

From the 43 PAM50 proteins we also removed proteins that had over 10% of values missing, leaving 33 proteins in this set. We performed k-Nearest Neighbor imputation with $k = 8$ [2] for the rest of the missing values in this case as well.

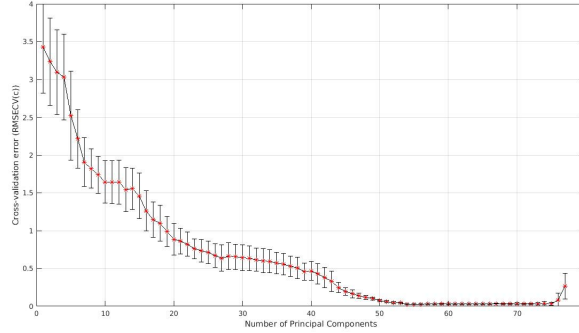
Figure 1: Percentage of values missing. Bars indicate number of proteins for which the percentage of patients with no data lies within the interval specified.



2.3 Principal Component Analysis

The large dataset, with 9313, is high-dimensional, which can be difficult to analyze. In order to reduce the dimensionality, we performed Principal Component Analysis (PCA) on the imputed dataset using the built-in MATLAB PCA function. To determine the optimal number of principal components we used 10-fold Cross-validation by Eigenvector. Figure 2 shows that at Principal Component (PC) number 54 the Cross-Validation Root Mean Square Error of (RMSECV) is minimized and 90.81 % of dataset variance is explained, suggesting that 54 is the optimal number of PCs to use for analysis.

Figure 2: 10-fold Cross-validation by Eigenvector shows that the optimal number of Principal Components is 54.



For the PAM50 dataset, PCA was not needed because it already has a low number of variables (33 proteins). We did mean center the data before clustering, something that is included in PCA for the larger dataset. Mean centering the data allows for more reliable comparisons between the proteins. Mean-centered data represents the difference of each value from the mean of the variable to which it belongs. Uncentered values may be misleading, as a particular protein with over-all higher values would seem more important than others.

2.4 Clustering

Using the scores for the 77 patients in the selected 54 PCs as coordinates, we clustered the data using three different methods: hierarchical; k-means; and fuzzy clustering. Two distance measures - squared Euclidean and cosine - were selected as parameters for the different clustering methods. Other distance measures were also tried, but the selected two provided the best results.

Hierarchical clustering with weighted average linkage was used to create dendrograms that help visualize clusters. Hierarchical clustering with weighted average linkage clusters the patients in a hierarchical manner, consecutively grouping the most similar patients together from a small cluster of the two most similar patients to a complete 77-patient cluster. Using different cutoff-values, a specific number of clusters can be specified.

K-means clustering was then performed with 3, 4 and 5 clusters. Because results may vary depending on the initial, randomly assigned, locations of the cluster centers, K-means clustering was performed many times for up to 10 000 iterations for the larger dataset until converging in a global minimum.

Fuzzy c-means clustering was also used to examine the data. Similar to k-means clustering, Fuzzy c-means requires the user to select the number of clusters (c). Different clusterings of 3, 4 and 5 clusters were performed. Each clustering was run until improvement in the MATLAB fuzzy means objective function was less than $1e-5$. The objective function in MATLAB fuzzy c-means represents euclidean distance. As output, Fuzzy c-means generates a membership grade between 0 and 1 specifying how well each point fits into the different clusters. Patients were assigned to clusters based on the highest grade given. Each clustering was run 5 times for 3,000 iterations in an effort to escape

problems with local minima. The exponent value for the fuzzy partition matrix was adjusted for the datasets and optimized based on silhouette output.

Adjusted Rand Indices (ARI) and Silhouette plots were used to assess the validity of each clustering. ARI is a measure of similarity between two data clusterings. It takes the original Rand Index and adjusts for chance events. The measure ranges from zero to one, with a chance for negative values in the case of unexpected clusterings. A value of one indicates that clusters match perfectly and a score of zero indicates that all similar clusterings are most likely due to chance event. Adjusted Rand Index has been extensively studied and is identified as one of the best methods for validation studies [7]. As "expert" clustering dataset we used the assigned subgroups based on PAM50-mRNA analysis. Silhouette plots provide an indication of the quality of each clustering. The silhouette shows which objects lie well within their cluster and which ones are somewhere in between clusters, as measured by the difference between their distance to points within the same cluster and those in other clusters [5]. To assess cluster quality, we look for the clustering with the highest Silhouette value.

3 Results

3.1 K-means clustering

3.1.1 Proteins selected based on PAM50 dataset

K-means clustering was run for 3000 iterations to give the algorithm a chance to meet a global minimum. Many optimal K-s were studied, but in the end the results for three, four and five clusters are shown for comparison.

Looking at the distance matrices for both distance measures (see Figure 3 & 4), it is clear to see that cosine distance provides a better separation of the clusters. Visual inspection shows the best separation is provided for three clusters.

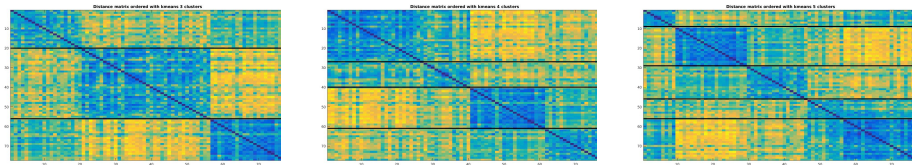


Figure 3: Distance matrices for K-means clustering with Cosine distance measure 3, 4 and 5 clusters

The highest ARI was achieved with Squared Euclidean distance measure and five clusters (see Table 1).

Silhouette plots (see Figure 5 & 6) reveal that the best distance measure and number of clusters combination is the Cosine distance measure with three clusters. K-means using three clusters and squared Euclidean distance also achieved good results compared to others. The Silhouette plots for these combinations show that only one or two objects have not been properly assigned to the spe-

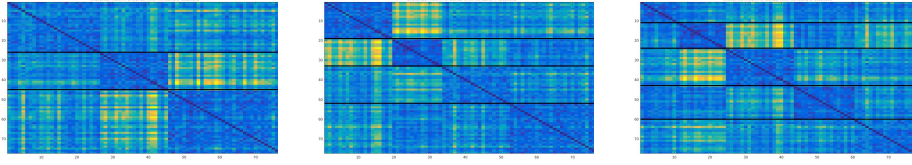


Figure 4: Distance matrices for K-means clustering with Squared Euclidean distance measure 3, 4 and 5 clusters

Table 1: Rand Indices for K-means clustering with Cosine and Squared Euclidean distance measures

Distance Measure	3 clusters	4 clusters	5clusters
Cosine	0.32	0.36	0.33
Squared Euclidean	0.36	0.30	0.42

cific cluster. Overall dividing objects into clusters is not that successful because looking at values for the objects they are mostly below 0.4.

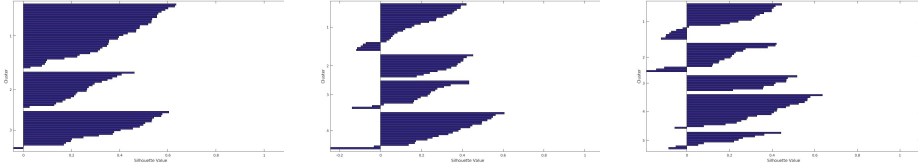


Figure 5: Silhouette plots for K-means clustering with Cosine distance measure for 3, 4 and 5 clusters

3.1.2 Whole dataset

The distance matrices (see Figures 7 & 8) reveal no good clustering option for this dataset.

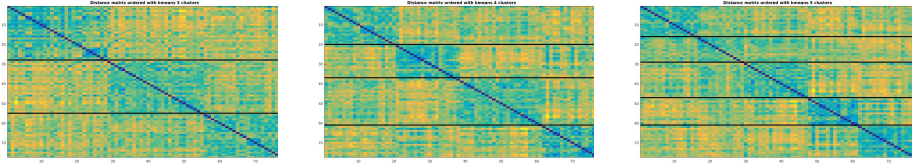


Figure 7: Distance matrices for K-means clustering with Cosine distance measure 3, 4 and 5 clusters

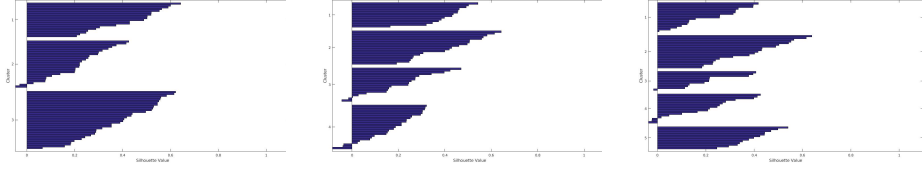


Figure 6: Silhouette plots for K-means clustering with Squared Euclidean distance measure for 3, 4 and 5 clusters

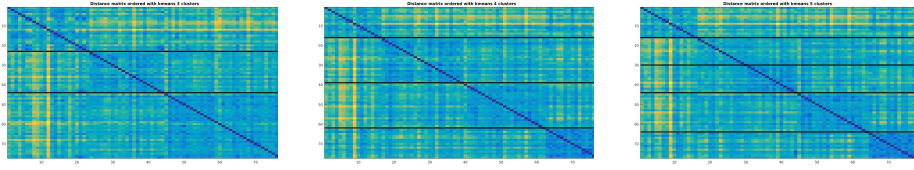


Figure 8: Distance matrices for K-means clustering with Squared Euclidean distance measure 3, 4 and 5 clusters

As shown in table 2, the indices obtained for K-means clustering are extremely low and indicate that the results of the clustering do not match labels assigned to patients based on TCGA microarray tests.

Table 2: Rand Indices for K-means clustering with Cosine and Squared Euclidean distance measures

Distance Measure	3 clusters	4 clusters	5clusters
Cosine	0.09	0.1	0.1
Squared Euclidean	0.16	0.1	0.17

Silhouette plots (see Figures 9 & 10) seem to suggest that the best clustering option is with squared Euclidean 3 clusters. Although this cannot be considered a particularly good clustering. Most of the objects have values lower than 0.2 indicating that the clustering is poorly done.

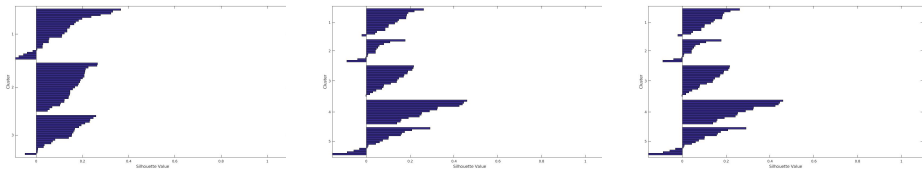


Figure 9: Silhouette plots for K-means clustering with Cosine distance measure for 3, 4 and 5 clusters

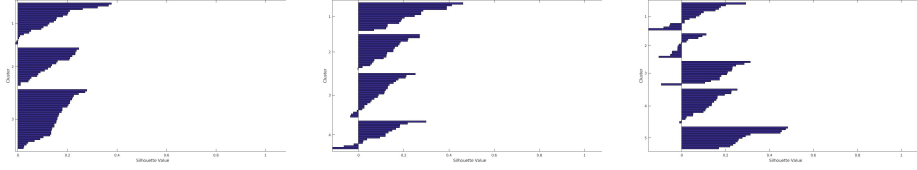


Figure 10: Silhouette plots for K-means clustering with Squared Euclidean distance measure for 3, 4 and 5 clusters

3.2 Fuzzy clustering

3.2.1 PAM50 proteins

The Adjusted Rand Index (ARI) for the small dataset compares the clusterings created with fuzzy to the "expert" dataset. ARI's range from 0.23 to 0.36, with the highest value a result of clustering with 3 clusters and using a fuzzy factor of 1.1 (Table 3). Values are slightly above random and show that there are only slight if any, similarities between the proteome clustering and the expert clustering. Silhouette values, in table 6, show that fuzzy clustering had the highest silhouette values with a fuzzy factor of 1.1.

3.2.2 Whole dataset proteins

Running fuzzy c-clustering on the expert dataset resulted in extremely low and variable Adjusted Rand Indices. All values are close to 0 or even negative indicating that any correlation between fuzzy clustering and "expert" clustering is random. Fuzzy was first run with a fuzzy factor of 2.0, the default value in MATLAB. This value was adjusted because cluster centers were too close and the algorithm couldn't separate the data into 4 clusters. The fuzzy factor was adjusted to get the best silhouette plot which occurred at the lowest possible fuzzy factor value of 1.1. The difference the fuzzy factor makes in clustering can be observed by looking at the silhouette plots in figure 11. On the large dataset, fuzzy clustering performed moderately compared to the other algorithms (Table 5).

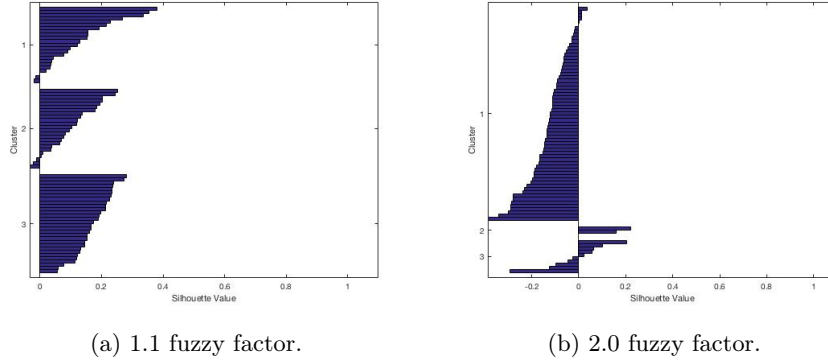


Figure 11: Silhouette plots. a) Silhouette plot with 3 clusters and a fuzzy factor of 1.1 b) compared with a plot with 3 clusters using a fuzzy factor of 2.0.

Table 3: ARI values using fuzzy clustering

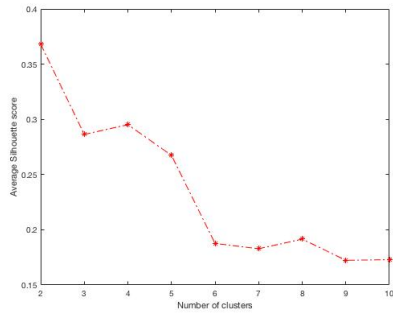
Dataset	Fuzzy Factor	3 clusters	4 clusters	5clusters
Small	1.1	0.3592	0.2951	0.2317
Small	2.0	0.2551	0.3000	0.2893
Large	1.1	0.1266	0.1186	0.0045
Large	2.0	-0.0043	-0.0110	0.0029

3.3 Hierarchical clustering

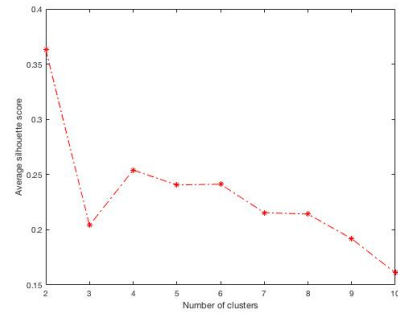
3.3.1 PAM50 proteins

Plots of the mean Silhouette values for hierarchical clusterings of the 33 PAM50 proteins, with squared Euclidean distance or cosine as distance measure, provide insight into the quality of different numbers of clusters. In both cases, two clusters give the highest Silhouette value, followed by four clusters (Figure 12).

Figures 13 and 14 show the clustering based on PAM50 proteins with squared Euclidean distance and cosine respectively. The score for clustering with squared Euclidean distance (Figures 13b and 13d) is slightly higher than with cosine as distance measure (Figures 14b and 14d).

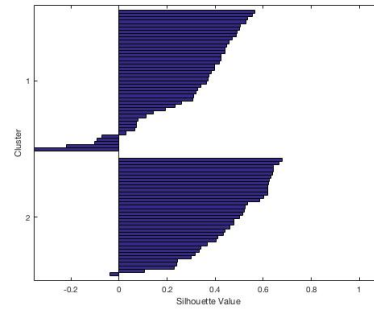
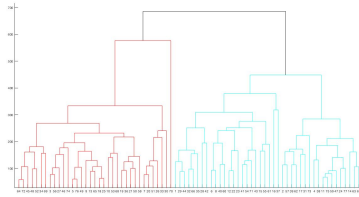


(a) Squared Euclidean.

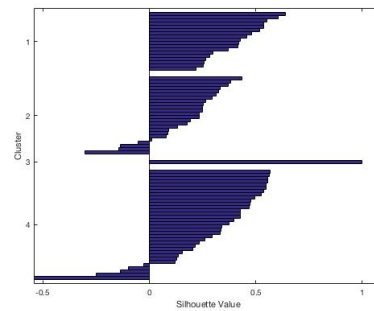
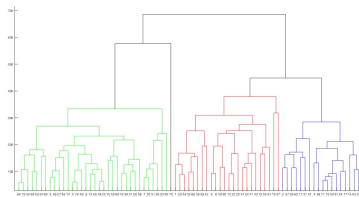


(b) Cosine.

Figure 12: Silhouette scores for hierarchical clustering based on PAM50 proteins, using two distance measures: squared Euclidean and cosine. Two clusters give the highest average score in both cases; four gives the second-highest average score.

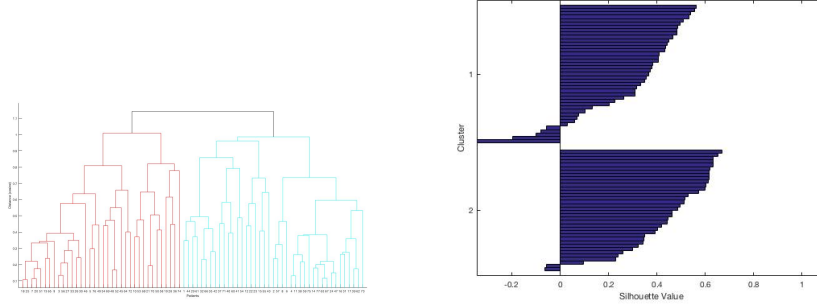


(a) Dendrogram with two clusters labeled. (b) Silhouette plot for two clusters. Average Silhouette score: 0.3682

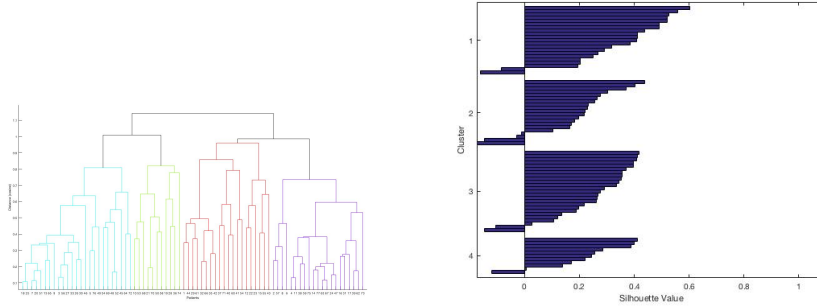


(c) Dendrogram with four clusters labeled. (d) Silhouette plot for four clusters. Average Silhouette score: 0.2952

Figure 13: Hierarchical clustering based on PAM50 proteins only, with squared Euclidean distance.



(a) Dendrogram with two clusters labeled. (b) Silhouette plot for two clusters. Average Silhouette score: 0.3630

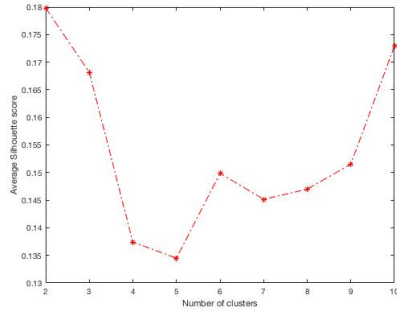


(c) Dendrogram with four clusters labeled. (d) Silhouette plot for four clusters. Average Silhouette score: 0.2540

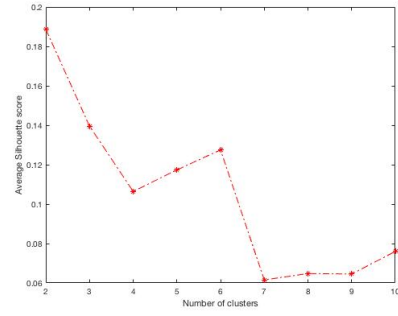
Figure 14: Hierarchical clustering based on PAM50 proteins only, with cosine as distance measure.

3.3.2 Whole dataset

Among the clusterings of the 9313 proteins in 54 dimensions, the highest score belongs to one with two clusters created using the cosine distance measure (Figure 17b). Again, plots of Silhouette values suggest that two clusters are optimal with both distance measures (Figures 16 and 17). In this case, six clusters give the second-highest scores (Figure 15); corresponding dendrograms and Silhouette plots are shown here as well (Figures 16c; 16d; 17c; 17d).

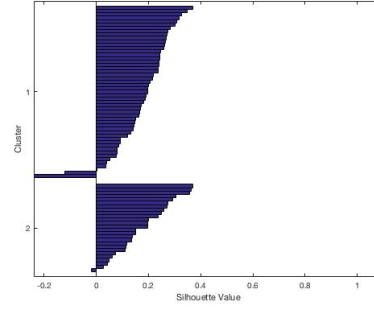
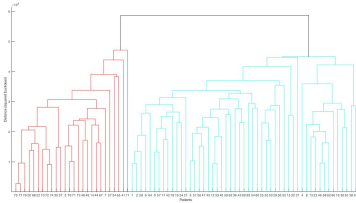


(a) Squared Euclidean.

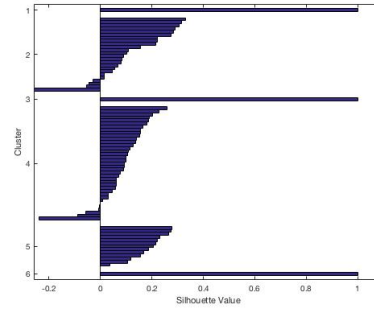
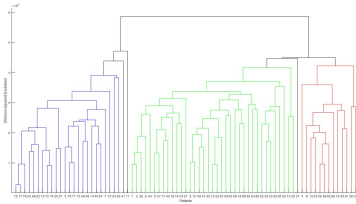


(b) Cosine.

Figure 15: Silhouette scores for hierarchical clustering based on 9313 proteins, reduced to 54 dimensions, using squared Euclidean distance and cosine. Two clusters give the highest score in both cases; a peak in average Silhouette scores is observed at 6 clusters.

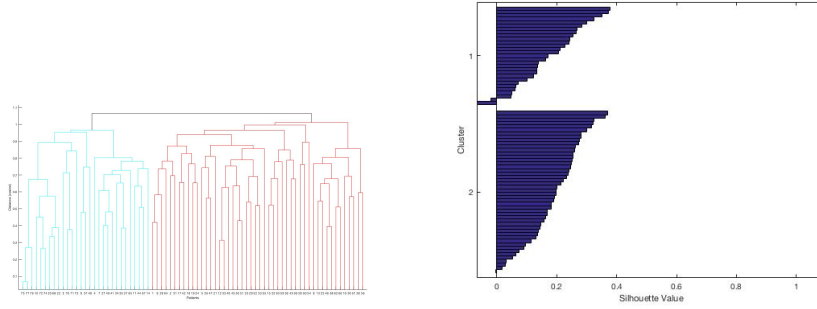


(a) Dendrogram with two clusters labeled. (b) Silhouette plot for two clusters. Average Silhouette score: 0.1798

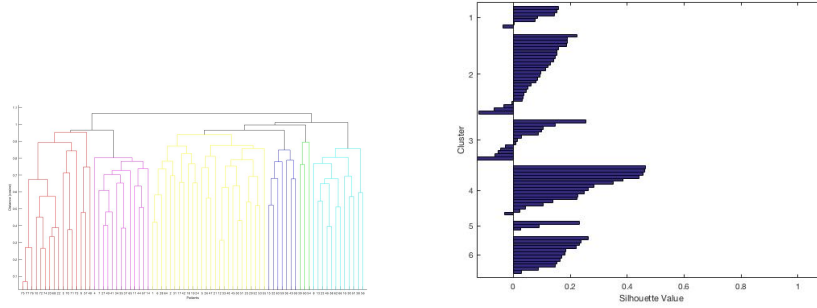


(c) Dendrogram with six clusters labeled. (d) Silhouette plot for six clusters. Average Silhouette score: 0.1499

Figure 16: Hierarchical clustering based on 9313 proteins, reduced to 54 dimensions, with squared Euclidean.



(a) Dendrogram with two clusters labeled. (b) Silhouette plot for two clusters. Average Silhouette score: 0.1887



(c) Dendrogram with six clusters labeled. (d) Silhouette plot for six clusters. Average Silhouette score: 0.1275

Figure 17: Hierarchical clustering based on 9313 proteins, reduced to 54 dimensions, with cosine as distance measure.

Table 4 shows the Adjusted Rand Index (ARI) for a range of clusterings of both datasets with squared Euclidean and cosine distance. These values indicate the degree of similarity to the grouping based on PAM50-mRNA analysis. All ARIs for clustering of the whole dataset are small negative values, indicating no similarity whatsoever to the PAM50 grouping. Using only the PAM50 proteins however, gives ARIs up to around 0.35, suggesting some similarity, as was expected from this select set of proteins.

Table 4: Adjusted Rand Indices (ARI) for hierarchical clusterings. Highest values for each dataset-distance measure combination in bold font.

Distance measure	Number of clusters	PAM50 clustering	All proteins
sqEuclidean	2	0.1771	-0.0082
sqEuclidean	3	0.1662	-0.0065
sqEuclidean	4	0.3505	-0.0086
sqEuclidean	5	0.3438	-0.0113
sqEuclidean	6	0.3028	-0.0177
Cosine	2	0.1659	-0.0047
Cosine	3	0.1508	-0.0091
Cosine	4	0.3145	-0.0044
Cosine	5	0.3124	-0.0156
Cosine	6	0.3236	-0.0070

3.4 ANOVA

Analysis of variance testing (ANOVA) was used to determine the significance of the Silhouette scores. For the large dataset, after running an ANOVA test ($\alpha = 0.05$) with a fixed number of clusters (4), we found that the means statistically differ from each other. The calculated p-value was smaller than the alpha and the hierarchical squared Euclidean was the best method with an average value of 0.1374. For the smaller dataset, k-means with 4 clusters using Euclidean was found to be significantly better than the other clusterings, with a silhouette value of 0.3093. It is important to note that the Silhouette values were calculated using Euclidean distances. As a result, Euclidean distance methods may appear more superior than they are in reality. Future calculations should be performed using multiple distances measures and averaged.

4 Discussion

4.1 Adjusted Rand Indices

The aim of this project is to divide breast cancer patients into subtypes, using different clustering methods and distance measures, according to their individual protein expression signature. First, we tried different clustering methods on the proteins that were selected from the larger dataset and that belong to the microarray PAM50 set. Then we tried clustering methods on the bigger proteome dataset. Our intentions were to see if clustering of proteomic data matched clusterings of microarray data.

In the end, patients were not clustered in the same groups that they were assigned using the "expert" dataset. Results were more promising for the small dataset where ARI were found to be slightly above random, in the range of 0.3 to 0.42. The highest ARI value of 0.42 was for K-means clustering using 5 clusters and squared Euclidean distance. Still comparing four microarray groups to five proteome groups is not very reliable and if anything it is a testament to how bad our clusters actually are. Clustering of the larger, more inclusive dataset performed even worse, achieving a maximum ARI of 0.17 using K-means

Table 5: Average Silhouette scores for 2-10 clusters of the large dataset in each of the algorithms.

Nr. of clusters	k-Means sqEuclidean	k-Means Cosine	Fuzzy FF=1.1	Fuzzy FF=1.2	Hierarchical sqEuclidean	Hierarchical Cosine
2	0.1838	0.1675	0.1812	0.1703	0.1798	0.1887
3	0.1360	0.1282	0.1406	0.1348	0.1682	0.1396
4	0.1341	0.1338	0.1336	0.1348	0.1374	0.1064
5	0.1152	0.1204	0.1184	0.1195	0.1345	0.1174
6	0.1243	0.1068	0.1320	0.1055	0.1499	0.1275
7	0.1105	0.1024	0.1045	0.1072	0.1452	0.0616
8	0.0833	0.0830	0.1134	0.1009	0.1470	0.0648
9	0.0768	0.0585	0.1110	0.0956	0.1515	0.0645
10	0.0946	0.0805	0.1076	0.0985	0.1729	0.0760

clustering with squared Euclidean distance. This value is close to zero and indicates that proteomic groups did not match microarray data defined groups.

While ARI did not indicate a correlation between proteomic and microarray data, it does not mean that our clusterings failed. After all clustering is an exploratory method performed to find new information and to gain a better understanding of a dataset. Adjusted Rand Indices show that, even though none of our clusterings are very similar to those based on mRNA, using only PAM50 proteins does bring us closer to this grouping.

4.2 Silhouette analysis

Silhouette plots assess clustering by calculating the distance a point is to the members of its own cluster compared to members of other clusters. The silhouette plots obtained did not indicate that our clusterings were very successful. For most methods a few points were assigned negative values and the rest were around 0.2, indicating some randomness in the clusterings.

For all methods the highest Silhouette score is obtained with only two clusters (Tables 6 and 5). This makes sense for a dataset that does not cluster well generally: the algorithms are able to divide the data into two groups, with an already very low Silhouette score, and each subsequent division is even more difficult, leading to even lower scores.

4.3 General discussion

Overall it appears that our data did not provide a clear clustering for the breast cancer patients. Research by Mertins *et al.* (2016) [4] on the same dataset found that four clusters were optimal for the smaller dataset, while for the whole protein dataset three clusters was best. Clustering proteomic data for breast cancer is difficult due to the limited availability of samples as well as high variability in terms of patient backgrounds and cancer prognosis. Additional challenges arise from the fact that cancer affects a multitude of genes at once and these need to be sorted through to find the significant cancer driving genes. Our study could benefit from more extensive preprocessing and more intensive filtering methods when looking at the initial dataset. Many of the patients in

Table 6: Average Silhouette scores for 2-10 clusters based on PAM50 proteins in each of the algorithms.

Nr. of clusters	k-Means sqEuclidean	k-Means Cosine	Fuzzy FF=1.1	Fuzzy FF=2.0	Hierarchical sqEuclidean	Hierarchical Cosine
2	0.3879	0.3554	0.3920	0.3920	0.3682	0.3630
3	0.3412	0.3023	0.3412	0.2551	0.2864	0.2040
4	0.3093	0.2821	0.2842	0.2214	0.2952	0.2540
5	0.2800	0.2539	0.2886	0.1869	0.2674	0.2406
6	0.2872	0.2390	0.2616	0.1807	0.1876	0.2413
7	0.2505	0.2144	0.2541	0.2259	0.1828	0.2155
8	0.2423	0.2372	0.2686	0.1594	0.1915	0.2142
9	0.2268	0.1915	0.2512	0.1064	0.1724	0.1919
10	0.1871	0.1715	0.2449	0.1009	0.1729	0.1610

this dataset were at varying stages in their cancer progression which may be reflected in their gene expression. Additionally, two male patients are included, whose genetic signatures could influence the clustering scheme. These cases may be eliminated, as well as other possible outliers. Ensemble methods such as consensus clustering can be performed to find the optimal clusterings.

The fact that proteomic and microarray clusterings did not match could lead to further investigation into new proteins that define these cancer types in the proteome. After all, not all genes or proteins that actually affect breast cancer have been identified, so that these kinds of methods that concentrate on searching for patterns in different cancer-related datasets are obviously highly appreciated. Moreover, it is likely that a discrepancy exists biologically between which genes are over- or underexpressed and useful for mRNA analysis and which proteins have increased or decreased levels in different cancer types.

In summary, using as many proteins as possible for clustering breast cancer patients is not useful. A set of genes, called PAM50, has been previously identified for grouping patients into four subtypes based on their mRNA levels. Clustering proteins that correspond to these genes gives clusterings of better quality, as well as higher ARIs. This suggests that this is a good start, although the quality still is too low to use in clinical applications. Future work should focus on increasing the number of samples, as well as identifying specific proteins that can be used for diagnosis instead of the PAM50 set. It should be noted that we only used 33 proteins that correspond to PAM50 genes, while many more exist (several per gene). This may contribute to the low quality of our clusterings; additional replicates in which more of these proteins are detectable could already greatly improve results.

5 Conclusion

Clustering as a method is exploratory and can lead to a better understanding of information and new solutions to existing problems. Clustering breast cancer patients based on a large set of proteomic data is not effective; clustering based on PAM50 proteins may be useful with further adjustments.

6 Supplementary material

All code and input files used in this project can be found on GitHub at <https://github.com/CharlottesNoort/ML-Project-Group56>.

7 Contributions

Table 7: Distribution of workload for the present project.

Name	Contribution
K. Ligi	20%
A. Gallagher	20%
C. van Noort	20%
I. Malloul	20%
A. Elezovic	20%

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